

AFYONKARAHISAR, Turkey

WMO: 171890

Lat: 38.733N

Lon: 30.600E

Elev: 3284

StdP: 13.03

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	16.1	6.5	8.7	13.4	11.2	11.0	18.7	25.0	43.7	20.6	41.2	2.8	310	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	92.3	62.7	89.4	62.2	86.6	61.6	66.0	84.3	64.7	82.6	63.6	81.1	7.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	88.1	72.2	58.6	83.0	71.1	57.1	78.5	70.1	32.6	83.9	31.5	82.4	30.6	81.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
19.8	16.7	14.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			2.9	96.2	9.6	3.3	-4.0	98.6	-9.7	100.5	-15.1	102.4	-22.1	104.7
			2.3	68.4	9.4	2.1	-4.5	69.9	-10.0	71.2	-15.2	72.3	-22.0	73.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	33.3	36.6	43.0	50.8	59.0	66.0	72.4	72.3	64.6	54.9	44.6	36.5	(6)
(7)		DBStd	15.23	9.33	8.94	7.68	7.19	5.95	5.16	4.38	4.54	5.62	6.76	7.62	8.42	(7)
(8)		HDD50	1860	519	379	238	73	6	0	0	0	32	190	423		(8)
(9)		HDD65	5030	983	795	683	428	200	48	2	4	74	318	611	884	(9)
(10)		CDD50	2924	1	3	20	97	284	481	694	691	437	183	29	4	(10)
(11)		CDD65	620	0	0	0	2	13	79	230	230	61	5	0	0	(11)
(12)		CDH74	7870	0	0	1	37	288	1110	2761	2653	937	83	0	0	(12)
(13)		CDH80	2932	0	0	0	2	42	329	1162	1108	282	7	0	0	(13)
(14)	Wind	WSAvg	5.3	5.0	5.3	5.9	5.6	5.3	5.8	6.3	5.8	5.2	4.5	4.3	4.6	(14)
(15)	Precipitation	PrecAvg	18.4	1.9	1.7	1.9	2.0	2.0	1.4	0.7	0.8	1.6	1.5	2.1		(15)
(16)		PrecMax	28.3	5.0	4.2	4.0	3.5	4.5	6.3	3.3	2.8	3.3	3.5	6.7		(16)
(17)		PrecMin	12.3	0.2	0.3	0.5	1.0	0.1	0.1	0.0	0.0	0.0	0.2	0.2		(17)
(18)		PrecStd	3.1	1.0	1.0	0.9	0.7	1.1	1.3	0.7	0.7	0.9	0.9	0.9	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	63.0	69.2	78.1	84.4	90.6	96.8	96.5	90.6	80.6	68.6	60.6	(19)
(20)		MCWB	45.0	47.9	49.4	55.3	60.6	61.8	62.8	64.1	60.1	57.7	51.6	46.8		(20)
(21)		2%	DB	51.7	56.8	64.5	73.5	80.4	87.2	92.7	92.6	86.4	76.1	64.3	54.9	(21)
(22)		MCWB	43.7	45.2	47.9	53.4	58.3	61.6	63.2	62.8	59.8	56.0	51.0	45.9		(22)
(23)		5%	DB	48.5	53.0	60.6	69.1	77.1	83.8	89.6	89.5	83.0	72.2	60.6	51.3	(23)
(24)		MCWB	42.1	43.5	46.3	52.0	57.4	60.7	62.6	62.2	58.8	54.7	49.3	44.6		(24)
(25)		10%	DB	45.5	49.5	56.7	64.8	73.7	80.6	86.6	86.2	79.6	68.3	57.1	48.3	(25)
(26)		MCWB	40.5	42.2	45.0	50.2	56.2	59.7	62.0	61.9	58.3	53.6	47.7	42.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	48.8	51.3	56.9	63.4	66.1	68.2	67.7	64.4	60.4	54.6	49.1	(27)
(28)		MADB	53.6	60.0	65.3	73.6	79.4	83.0	87.3	85.6	83.4	75.8	64.1	56.7		(28)
(29)		2%	WB	44.0	46.0	49.0	54.5	60.3	63.9	66.1	66.0	62.3	57.9	51.9	46.7	(29)
(30)		MADB	50.7	55.2	61.5	70.1	75.8	80.8	85.3	84.0	79.4	70.7	61.6	53.3		(30)
(31)		5%	WB	42.3	44.1	47.2	52.8	58.6	62.4	64.7	64.8	60.9	56.4	50.1	45.0	(31)
(32)		MADB	47.7	52.0	58.7	66.7	73.5	78.7	83.1	82.4	77.2	68.2	58.8	50.8		(32)
(33)		10%	WB	40.7	42.3	45.5	51.0	57.0	61.1	63.5	63.6	59.6	54.9	48.4	43.2	(33)
(34)		MADB	45.1	48.8	55.4	63.3	71.1	76.9	81.6	80.9	75.2	65.9	55.8	47.8		(34)
(35)	Mean Daily Temperature Range	MDBR	13.4	15.9	18.6	20.7	23.6	25.1	26.9	26.6	26.3	21.9	19.7	13.7	(35)	
(36)		5% DB	MCDBR	17.1	21.9	26.2	28.1	29.7	31.0	31.0	31.3	32.6	28.7	25.1	18.9	(36)
(37)		MCWBR	10.8	12.5	13.2	12.6	12.6	11.0	9.7	9.4	12.4	13.1	13.7	11.9	(37)	
(38)		5% WB	MCDBR	15.6	20.0	23.1	24.9	27.1	26.6	27.1	26.5	27.0	23.8	21.4	16.7	(38)
(39)		MCWBR	10.2	11.9	12.4	12.1	12.5	10.9	9.8	8.9	11.0	11.8	12.2	10.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.320	0.364	0.394	0.380	0.359	0.355	0.358	0.338	0.335	0.307	0.294	(40)	
(41)		taud	2.424	2.370	2.267	2.210	2.309	2.388	2.393	2.366	2.422	2.429	2.474	2.477	(41)	
(42)		Ebn at Noon	281	286	283	280	285	291	291	288	288	278	274	273	(42)	
(43)		Edh at Noon	26	32	39	44	41	38	37	37	33	30	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	646	929	1301	1669	1982	2283	2413	2140	1702	1186	828	582	(44)	
(45)		RadStd	73	125	101	134	149	171	88	75	113	78	85	83	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	-0.94	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page